Executive Business Report

End- to- End Sales Forecasting & Demand Intelligence System

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Executive Summary

This project was developed to analyze historical Superstore sales data and provide accurate sales forecasting for future business planning. The system combines data analysis, time series forecasting, anomaly detection, product demand segmentation, and an interactive Streamlit dashboard into one complete solution.

The dataset was cleaned and preprocessed before analysis. Three forecasting models—SARIMA, Facebook Prophet, and XGBoost—were implemented and compared using evaluation metrics such as MAE, RMSE, and MAPE. The best-performing model was selected to forecast sales for the next three months.

The project also identifies unusual sales spikes and drops using anomaly detection techniques and segments products into different demand groups using K-Means clustering. These insights help business managers make informed decisions regarding inventory management, stock planning, and demand forecasting.

Key Findings:

1. Exploratory Data Analysis (EDA)

• Sales data was successfully cleaned and preprocessed.

• Date columns were converted into proper datetime format.

• Monthly and yearly sales trends were generated.

• Seasonal sales patterns were identified.

• Sales performance was analyzed by category and region.

2. Sales Forecasting

Three forecasting models were implemented:

• SARIMA

• Facebook Prophet

• XGBoost

The models were compared using MAE, RMSE, and MAPE values. The model with the lowest prediction error was selected as the final forecasting model.

3. Three-Month Sales Forecast

The selected forecasting model predicts stable sales growth over the next three months. The forecast indicates that demand is expected to remain steady with minor seasonal fluctuations.

The confidence interval suggests that actual sales are expected to remain within the predicted range under normal business conditions.

4. Anomaly Detection

• Isolation Forest and Z-Score techniques were used to detect unusual sales behavior.

• Possible reasons behind anomalies include:

• Festival season sales

• Promotional offers

• Flash discounts

• Sudden customer demand

• Inventory shortages

These anomalies provide valuable insights for future planning and risk management.

5. Product Demand Segmentation

K-Means clustering grouped products into different demand categories such as:

• High Demand

• Stable Demand

• Low Demand

• Growing Demand

This segmentation enables efficient inventory management and improves product stocking decisions.

Business Recommendations:

Recommendation 1

Increase inventory for products that consistently belong to the High Demand cluster to avoid stock shortages.

Recommendation 2

Monitor anomalies regularly to identify unexpected sales spikes or drops and respond quickly with operational actions.

Recommendation 3

Use the selected forecasting model for monthly inventory planning, purchasing decisions, and demand forecasting to reduce overstocking and understocking.

Limitation:

The forecasting system is based on historical sales data. Unexpected market changes, customer behavior, economic conditions, or special promotional events may affect prediction accuracy. Therefore, the forecasting model should be updated periodically using the latest sales data.

Conclusion:

The End-to-End Sales Forecasting and Demand Intelligence System successfully combines data analysis, forecasting, anomaly detection, clustering, and dashboard visualization into one business solution.

The project helps organizations improve inventory management, reduce operational costs, forecast future demand, and support data-driven decision-making. The interactive Streamlit dashboard further enhances usability by allowing business users to explore sales trends and forecasting results without requiring programming knowledge.